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CO1-AT2

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1.

Operating System

1. OS TYPES

- Batch OS
- Time Sharing
- Distributed
- Real-time OS
- Network OS

Use/ based on →

2. OS STRUCTURES

- Monolithic Structure
- Layered Structure
- Modular Structure

determine organization →

3. SYSTEM CALLS

- Process Calls
- File Calls
- Memory Calls
- Device Calls
- Information Calls

enable access to →

4. OS FUNCTIONS

- Process Management
- Memory Management
- File Management
- Device Management
- Security

Different OS types use different structures

OS Structure influences how System calls are implemented

System call provide access to OS functions

Different OS types emphasize different OS Functions

(e.g., Real-Time OS → Fast Response)

5.

SYSTEM CALLS

1. Process System Calls

- Create process
- Terminate process
- Load / Execute
- Wait / Signal
- Get/Set attribute

2. File System Calls

- Create file
- Delete file
- Open/close file
- Read/write file
- Get / Set file

3. Device System Calls

- Request device
- Release device
- Read from device
- Write to device
- Get / Set device

4. Information System Calls

- Get System time
- Get system data
- Get process / file
- Set system data
- Get / Set attribute

Relation to OS Services

A. Scheduling (CPU Management)

Process System Calls are used by OS to schedule & allocate CPU

B. Memory Allocation

Process & information system calls need memory allocation by OS.

C. Device Handling (I/O Management)

Device System Calls are used by OS for I/O & device handling

D. Information Services

Information System calls provide OS information services to users